Divergence

Natural selection simulation — Quick reference

Getting started: open the hosted link, pick a starting environment (Mild world, Cold gradient, Toxic basin, or Rapid evolution), and click Start run. Use Space to pause, 1–4 to change speed, click any cell to inspect its traits and brain, or trigger events from the bottom bar.

Overview

Divergence simulates natural selection in continuous 2D space. Cells move, eat food pellets, avoid drifting toxin, and cope with local temperature. Individuals better suited to their surroundings survive longer and leave more offspring, shifting population traits over generations until lineages split into distinct species.

Controls

Drag anywhere in the water to pan and scroll to zoom around the cursor. Click any living cell to inspect its traits, energy, sense radius, and live neural outputs. Space pauses; 1–4 set speed; T toggles thermal overlay; F resets view; Tab hides charts; Escape cancels or closes.

World setup

A seed fixes the initial terrain and starting population so identical settings always begin from the same world. Presets configure ready-made environments, or you can adjust food abundance, temperature contrast, climate scale, starting toxin, and mutation rate manually.

Environment

Temperature varies smoothly across the water from cool blue currents to warm coral regions. Food spawns more densely in fertile patches, while toxin particles drift, attract slightly, and gradually decay.

Inherited traits

Every cell inherits ten numerical traits (size, speed, sense range, preferred temperature, temperature tolerance, toxin resistance, diet, reproduction threshold, sexual tendency, and choosiness) plus a 100-weight neural brain. Each trait carries an energy or survival trade-off.

Reproduction

Cells with sexual tendency below 0.50 divide on their own once they store enough energy. At 0.50 or above, two compatible neighbours within sense range must both have sufficient energy to combine genomes, with small random mutations applied at birth.

Species & colour

Founders begin near white. As mutations accumulate in traits and brain weights, genome colours shift smoothly in perceptual colour space. Grazers show a green inner organelle, while high-diet hunters develop warm amber cores and front mandibles.

Reading charts

Traits plots mean ± 1 SD (dashed line marks founders) and value bands against time or mean generation. Species stacks the top 20 lineages by peak size; click any living species to spotlight it. Tree shows speciation splits; Stats tracks live rates and causes of death.

Saving & data

Save downloads a JSON snapshot that resumes identically when loaded. Export CSV downloads trait time-series and species population tables for plotting or lab analysis. Save PNG captures the window.

Environmental events

Heat wave	Temporarily raises local temperature before easing back to baseline.
Ice age	Temporarily lowers local temperature before easing back to baseline.
Famine	Clears most food in the target area and slows regrowth while active.
Food bloom	Spawns dense extra food across the target area while active.
Toxin spill	Releases a cloud of toxin particles that slowly disperses and decays.

Minimum window size: 1280 × 720. Supported in recent Chrome, Edge, Firefox, and Safari.